Mnemosyn

Reviewing the anatomy and clinical content of a student atlas

A booklet for specialist reviewers

October 2026, second edition

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What changed in this edition

This is the second edition of the booklet. The questions are the same, with the same numbers and the same wording, so the review form is unchanged and any answers you have already given still count.

Since the first edition the atlas has gained the spinal cord, with its segments, grey matter and tracts, and the brainstem nuclei of three further atlases. Several route stations that the first edition called words only are now shown in three dimensions: the lateral corticospinal tract, the dorsal columns, the dorsal horn, the periaqueductal grey, the pedunculopontine and laterodorsal tegmental nuclei, and the nucleus of the solitary tract. The notes on the motor, touch, pain, acetylcholine, fear and hunger routes now say so.

Some of these shapes come from atlases whose authors have not yet allowed sharing. They appear in the project's own builds only, and in shared copies of the atlas these stations are described in words until permission arrives.

1. A letter to the reviewer

Dear colleague,

Mnemosyn is a free atlas of the human brain and spinal cord made for students. It lets them turn the brain in three dimensions, cut through it in thin slices, follow the great sensory, motor and behavioural pathways station by station, and read what each structure does and what happens when it is damaged.

Before students learn from it, we want people who know this field to tell us whether we have it right. This booklet gathers everything we are not yet sure enough to teach: descriptions of structures, the routes of the main pathways, the brain circuits behind behaviour, the timing of a few hormone rhythms, and the handful of statements that gave us the most doubt.

For each topic we explain what we wrote, why we hesitate, and which textbooks and papers we relied on. At the end of each topic there is a numbered question. The review form asks the same numbered questions, so you can read here and answer there, at your own pace. There are fifty questions; most reviewers will need about an hour, and the form lets you come back to change your answers later.

You do not need any technical knowledge. We ask only for your judgment as a clinician or scientist. Every answer helps, and "outside my expertise" is a useful answer too.

Thank you for giving us your time.

2. How we wrote this content

Every description in this booklet was drafted from standard textbooks of neuroscience and from published research papers, and each statement names the source it rests on. What has not happened yet is the step that matters most: nobody has checked these statements against the exact pages of those sources, and no clinician has read them. Students will take what they read in an atlas as settled, so a wrong side of a deficit, a misplaced structure or an outdated claim would be learned as fact.

Three kinds of decisions need your eyes. The first are plain facts: what a structure does, what a lesion causes, which disorders and medicines are linked to a circuit. The second are simplifications. To let a student follow a pathway, we had to turn rich and branching anatomy into a short, ordered journey, and every such journey leaves something out. We tell you what we left out and why, and ask whether the result still teaches the right lesson. The third are classifications: where a structure belongs when textbooks group it in different ways.

When a question asks whether something is correct, please judge it as teaching material for medical, neuroscience and psychology students, not as a complete clinical reference.

How to answer

Please give your answers in the review form whose link came with this booklet.

- Correct as written: you would teach it this way.
- Correct after small changes: the idea is right but a detail, a word or a side needs fixing. Please write the change in the notes.
- Incorrect: the statement or choice is wrong. Please write what it should say.
- Outside my expertise or not sure: you prefer not to judge this one.

3. Seven structures

We begin with seven structures whose descriptions are written. Each has up to three parts: what it does, how it is connected, and what happens when it is damaged. The clinical part of the locus coeruleus is still missing; what we plan to write there is the subject of question 12. Please read each account as a whole and tell us whether it is accurate.

Precentral gyrus

What it does

Contains the primary motor cortex, which sends the commands for voluntary movement of the opposite side of the body (Purves et al. 2018). Body parts are represented in order along the gyrus, with the face and hand occupying a large share of it (Penfield and Boldrey 1937).

How it is connected

Receives input from premotor cortex, the supplementary motor area, primary somatosensory cortex and the ventral lateral nucleus of the thalamus, which relays output of the cerebellum and basal ganglia (Kandel et al. 2021). Its output descends in the corticospinal and corticobulbar tracts to the spinal cord and brainstem motor nuclei (Purves et al. 2018).

When it is damaged

Damage causes weakness of the opposite side of the body with upper motor neuron signs, such as spasticity and increased reflexes that develop over time (Purves et al. 2018).

Question 1. Precentral gyrus

Is this account of the precentral gyrus accurate?

□ Correct as written
□ Incorrect
□ Correct after small changes
□ Outside my expertise or not sure

Hippocampus

What it does

Required for forming new declarative memories of facts and events, and important for spatial memory (Kandel et al. 2021).

How it is connected

Its main cortical input comes from the entorhinal cortex through the perforant path; its output leaves mainly through the fornix and back to the entorhinal cortex (Purves et al. 2018).

When it is damaged

Bilateral removal of the medial temporal lobes, including the hippocampi, caused a severe and lasting inability to form new memories in the patient H.M. (Scoville and Milner 1957). The neurofibrillary changes of

Alzheimer disease appear early in the entorhinal region and hippocampus (Braak and Braak 1991).

Question 2. Hippocampus

Is this account of the hippocampus accurate?

□ Correct as written
□ Incorrect
□ Correct after small changes
□ Outside my expertise or not sure

Lateral geniculate nucleus

What it does

The thalamic relay for vision: it receives retinal input and passes it to the primary visual cortex (Purves et al. 2018).

How it is connected

Receives axons of retinal ganglion cells from both eyes through the optic tract, with the input from each eye kept in separate layers, and projects through the optic radiation to the primary visual cortex around the calcarine sulcus (Purves et al. 2018).

When it is damaged

Damage to one lateral geniculate nucleus causes loss of vision in the opposite visual field of both eyes (Kandel et al. 2021).

Question 3. Lateral geniculate nucleus

Is this account of the lateral geniculate nucleus accurate?

□ Correct as written
□ Incorrect
□ Correct after small changes
□ Outside my expertise or not sure

Substantia nigra, pars compacta

What it does

Contains dopamine-producing neurons whose projections to the striatum support the initiation and selection of movement through the basal ganglia circuits (Purves et al. 2018).

How it is connected

Sends dopaminergic axons to the caudate nucleus and putamen (the nigrostriatal pathway) and receives input from the striatum (Kandel et al. 2021).

When it is damaged

Degeneration of these dopaminergic neurons underlies Parkinson disease, with slowness of movement, rigidity and resting tremor (Kandel et al. 2021).

Question 4. Substantia nigra, pars compacta

Is this account of the substantia nigra, pars compacta accurate?

□ Correct as written
□ Incorrect
□ Correct after small changes
□ Outside my expertise or not sure

Locus coeruleus

What it does

The main source of noradrenaline (norepinephrine) for the forebrain, cerebellum and spinal cord; its activity is linked to arousal, attention and the sleep-wake cycle (Purves et al. 2018).

How it is connected

A small nucleus in the rostral pons near the floor of the fourth ventricle, whose highly branched axons reach most of the central nervous system (Kandel et al. 2021).

When it is damaged

We have not written this part yet.

Question 5. Locus coeruleus

Is this account of the locus coeruleus accurate?

□ Correct as written
□ Incorrect
□ Correct after small changes
□ Outside my expertise or not sure

Pituitary gland

What it does

The master gland of the endocrine system. Its anterior lobe releases growth hormone, prolactin, thyroid-stimulating hormone, adrenocorticotropic hormone and the gonadotropins under the control of hypothalamic releasing and inhibiting hormones; its posterior lobe releases oxytocin and vasopressin made in the hypothalamus (Purves et al. 2018).

How it is connected

Sits in the sella turcica below the hypothalamus, joined to it by the pituitary stalk. Hypothalamic hormones reach the anterior lobe through the hypophyseal portal veins; axons from the supraoptic and paraventricular nuclei run down the stalk and end in the posterior lobe (Kandel et al. 2021).

When it is damaged

A pituitary adenoma can press upward on the optic chiasm and take both temporal visual fields (bitemporal hemianopia), and can cause too much or too little of its hormones; loss of vasopressin release causes diabetes insipidus (Kandel et al. 2021).

Question 6. Pituitary gland

Is this account of the pituitary gland accurate?

□ Correct as written
□ Incorrect
□ Correct after small changes
□ Outside my expertise or not sure

Pineal gland

What it does

Makes melatonin, mostly at night, under the control of the light-dark cycle, and so helps set the daily sleep-wake rhythm (Purves et al. 2018).

How it is connected

A small gland on the roof of the third ventricle, behind the habenula and above the superior colliculi. Light information reaches it from the retina through the suprachiasmatic nucleus of the hypothalamus and a sympathetic pathway through the superior cervical ganglion (Purves et al. 2018).

When it is damaged

Tumours of the pineal region can press on the dorsal midbrain and cause Parinaud syndrome, with loss of upward gaze, and can block the cerebral aqueduct and cause hydrocephalus. The gland often calcifies with age, which shows on CT scans (Kandel et al. 2021).

Question 7. Pineal gland

Is this account of the pineal gland accurate?

□ Correct as written
□ Incorrect
□ Correct after small changes
□ Outside my expertise or not sure

4. Where structures belong

The atlas arranges every structure in a family tree: brain, forebrain, cerebral hemisphere, lobe and so on. Students browse this tree and are quizzed on whole families, so the tree teaches a classification. For most structures the choice is obvious. For the five below, textbooks disagree, or our choice was driven by the shape of the data we built on, and we would like your judgment.

The amygdala

We currently file the amygdala with the basal nuclei, next to the caudate, putamen and globus pallidus. We did so because the brain map we built on, the AAL3 atlas (Rolls et al. 2020), groups it among the deep grey nuclei of the hemisphere, and because classical anatomy lists the amygdaloid body among the basal nuclei of the cerebrum.

Most teaching, however, presents the amygdala as part of the limbic system, alongside the hippocampus and cingulate cortex, because of its central role in emotion and fear (Kandel et al. 2021; Purves et al. 2018). Others place it simply in the medial temporal lobe, where students meet it in slices. Each choice teaches something different.

Question 8. The amygdala

Is filing the amygdala with the basal nuclei acceptable for teaching? If not, please tell us where you would place it (limbic structures, medial temporal lobe, or elsewhere).

□ Correct as written
□ Incorrect
□ Correct after small changes
□ Outside my expertise or not sure

The olfactory cortex

We file the olfactory cortex in the frontal lobe, because in the brain map we build on (Rolls et al. 2020) it is drawn on the orbital surface beneath the frontal lobe.

Many textbooks instead describe it as old cortex (paleocortex) that belongs to the limbic or rhinencephalic structures rather than to any lobe, and some place its main part, the piriform cortex, at the junction of the frontal and temporal lobes (Purves et al. 2018).

Question 9. The olfactory cortex

Is the frontal lobe an acceptable home for the olfactory cortex in a student atlas? If not, where should it go?

□ Correct as written
□ Incorrect
□ Correct after small changes
□ Outside my expertise or not sure

The pituitary gland

We place the whole pituitary gland under the diencephalon, beside the hypothalamus. Our reason is that the posterior lobe is a downgrowth of the hypothalamus: it holds the endings of supraoptic and paraventricular neurons (Kandel et al. 2021).

The anterior lobe, however, is glandular tissue of a different origin and is not brain tissue at all. Some would rather show the pituitary as its own endocrine organ, outside the brain's family tree, or split it into its two lobes.

Question 10. The pituitary gland

Placing the whole pituitary gland under the diencephalon is acceptable for teaching. If you disagree, please tell us in the notes whether it should stand apart as an endocrine gland or be split into its two lobes.

□ Correct as written
□ Incorrect
□ Correct after small changes
□ Outside my expertise or not sure

The pineal gland

We created a small group called the epithalamus, under the diencephalon, and placed the pineal gland in it. Later the habenula will join it.

This follows the classical division of the diencephalon into thalamus, hypothalamus, subthalamus and epithalamus (Purves et al. 2018). We would like to know whether this is how you would teach it.

Question 11. The pineal gland

Is the epithalamus, within the diencephalon, the right place for the pineal gland?

□ Correct as written
□ Incorrect
□ Correct after small changes
□ Outside my expertise or not sure

A missing clinical note on the locus coeruleus

Our description of the locus coeruleus explains its function and connections but has no clinical part yet, because we were not sure what a student most needs to know.

We plan to say that it is among the earliest sites of change in Alzheimer's disease and is damaged in Parkinson's disease, and that its overactivity is linked to anxiety and to the hyperarousal of post-traumatic stress (Kandel et al. 2021; Aston-Jones and Cohen 2005).

Question 12. A missing clinical note on the locus coeruleus

Is this the right content for a clinical note on the locus coeruleus? Please add or correct anything in the notes.

□ Correct as written
□ Incorrect
□ Correct after small changes
□ Outside my expertise or not sure

5. Sensory and motor pathways

Each of the following routes is told as a journey from where a signal starts to where it ends. For every station we say what it does and what a lesion there causes. The atlas lets a student begin on either side of the body; here every route is told for a start on the left. The sides of later stations and of each deficit follow from that start, and the atlas swaps them when a student starts on the right. After each journey we describe the simplifications we made. The Papez circuit is told here too: it is not a sensory or motor pathway, but like them it runs on one side of the brain.

Visual pathway

It starts in the left eye.

Light falling on the retina is turned into signals that run through the optic nerve, partly cross at the optic chiasm, relay in the lateral geniculate nucleus of the thalamus and reach the primary visual cortex around the calcarine sulcus. Each hemisphere sees the opposite half of the visual field (Purves et al. 2018).

The journey

1. Optic nerve, left side. Carries the output of the retinal ganglion cells of the left eye to the optic chiasm (Purves et al. 2018). *If it is damaged:* Blindness of the left eye, with loss of its direct pupillary light reflex (Kandel et al. 2021).

2. Crossing at the optic chiasm. Fibres from the nasal half of each retina cross here, so each optic tract carries the opposite half of the visual field from both eyes; fibres from the temporal half of the left retina stay on the left side (Purves et al. 2018). *If it is damaged:* A midline lesion here, for example a pituitary tumour, cuts the crossing fibres and takes both temporal visual fields (bitemporal hemianopia) (Kandel et al. 2021).

3. Lateral geniculate nucleus, left side. The left lateral geniculate nucleus relays the right half of the visual field from both eyes, keeping each eye in separate layers (Purves et al. 2018). *If it is damaged:* Loss of the right half of the visual field in both eyes (right homonymous hemianopia) (Kandel et al. 2021).

4. Optic radiation, left side. Carries visual signals from the left lateral geniculate nucleus to the left visual cortex; fibres for the upper field loop forward through the temporal lobe (Meyer's loop) (Purves et al. 2018). *If it is damaged:* Loss of the right half of the visual field in both eyes; a temporal lobe lesion of Meyer's loop takes mainly the right upper quarter (Kandel et al. 2021).

5. Calcarine cortex, left side. The primary visual cortex maps the right half of the visual field, with central vision at the occipital pole (Purves et al. 2018). *If it is damaged:* Loss of the right half of the visual field in both eyes, often with central vision spared when the occipital pole keeps its blood supply (Kandel et al. 2021).

What we simplified

We show the lateral geniculate nucleus, the optic radiation and the visual cortex on the side of the starting eye. These carry the opposite half of the visual field from both eyes, which the text explains at each station.

Question 13. Visual pathway

Is our account of this pathway accurate: the order of the stations, their sides, the role and the effect of a lesion at each, and the simplifications we made?

□ Correct as written
□ Incorrect
□ Correct after small changes
□ Outside my expertise or not sure

Auditory pathway

It starts in the left ear.

Sound moves the hair cells of the cochlea, whose signals travel through the cochlear nerve to the cochlear nuclei, then up both sides of the brainstem through the superior olivary complex, the lateral lemniscus and the inferior colliculus to the medial geniculate nucleus and the primary auditory cortex. Because the pathway crosses early and on both sides, each hemisphere hears both ears (Purves et al. 2018).

The journey

1. Vestibulocochlear nerve, left side. The cochlear part of the left vestibulocochlear nerve carries sound signals from the hair cells of the left cochlea to the cochlear nuclei (Purves et al. 2018). *If it is damaged:* Hearing loss in the left ear, often with tinnitus; a vestibular schwannoma here also causes imbalance (Kandel et al. 2021).

2. Cochlear nuclei, left side. The left cochlear nuclei receive only the left ear, so this is the last station where each ear is heard alone (Purves et al. 2018). *If it is damaged:* Deafness in the left ear; this is the only level of the brain where a one-sided lesion causes one-sided deafness (Kandel et al. 2021).

3. Crossing at the trapezoid body. Most fibres from the cochlear nuclei cross the pons here toward the right superior olivary complex and lateral lemniscus, while some stay on the left side, so from here on each side carries both ears (Purves et al. 2018). *If it is damaged:* Damage here mainly impairs locating sounds, because the superior olivary complex compares the timing and loudness at the two ears (Purves et al. 2018).

4. Lateral lemniscus, right side. Carries signals from both ears, mostly the left one, up the right brainstem to the inferior colliculus (Purves et al. 2018). *If it is damaged:* A one-sided lesion above the cochlear nuclei rarely causes deafness, because each side carries both ears; locating sounds can suffer (Kandel et al. 2021).

5. Inferior colliculus, right side. The right inferior colliculus integrates differences in timing and loudness between the ears and maps where a sound comes from (Purves et al. 2018). *If it is damaged:* Usually mild difficulty locating sounds; lesions on both sides can cause marked hearing loss (Kandel et al. 2021).

6. Medial geniculate nucleus, right side. The right medial geniculate nucleus is the thalamic relay for hearing and projects to the primary auditory cortex (Purves et al. 2018). *If it is damaged:* Little or no hearing loss from one side, with difficulty locating sounds and following speech in noise (Kandel et al. 2021).

7. Acoustic radiation, right side. Carries auditory signals from the right medial geniculate nucleus to the right Heschl gyrus (Purves et al. 2018). *If it is damaged:* No deafness from one side, but poorer locating of sounds coming from the left side (Kandel et al. 2021).

8. Heschl gyrus, right side. The primary auditory cortex, mapped by sound frequency (Purves et al. 2018). *If it is damaged:* A one-sided lesion does not cause deafness, because each cortex hears both ears; it impairs locating sounds on the left side. Lesions on both sides cause cortical deafness (Kandel et al. 2021).

What we simplified

After the trapezoid body we draw the route on the opposite side only, although hearing rises on both sides of the brainstem. The text says so, but the picture a student sees is one-sided.

Question 14. Auditory pathway

Is our account of this pathway accurate: the order of the stations, their sides, the role and the effect of a lesion at each, and the simplifications we made?

□ Correct as written
□ Incorrect
□ Correct after small changes
□ Outside my expertise or not sure

Papez circuit

It starts in the left hippocampus.

In 1937 James Papez proposed a loop from the hippocampus through the fornix, the mammillary body, the anterior thalamus, the cingulate cortex and the parahippocampal gyrus back to the hippocampus as the basis of emotion (Papez 1937). Today it is seen mainly as a memory circuit: damage at several of its stations causes amnesia (Kandel et al. 2021).

The journey

1. Hippocampus, left side. The hippocampus binds new experiences into lasting memories and sends its output through the fornix (Purves et al. 2018). *If it is damaged:* One-sided damage causes a mild memory deficit, for words with a left lesion and for places and pictures with a right one; damage on both sides prevents new memories from forming (Scoville and Milner 1957).

2. Fornix, left side. The main output bundle of the hippocampus; it arches under the corpus callosum to the mammillary body (Purves et al. 2018). *If it is damaged:* Damage to both fornices, for example during surgery for a colloid cyst of the third ventricle, can cause lasting difficulty forming new memories (Kandel et al. 2021).

3. Mammillary body, left side. A hypothalamic nucleus that receives the fornix and relays to the anterior thalamus through the mammillothalamic tract (Purves et al. 2018). *If it is damaged:* Damaged on both sides in thiamine deficiency (Wernicke-Korsakoff syndrome), with severe memory loss and confabulation (Kandel et al. 2021).

4. Mammillothalamic tract, left side. Carries signals from the mammillary body to the anterior nuclei of the thalamus (Purves et al. 2018). *If it is damaged:* Thalamic strokes that cut this tract can cause memory loss like that of mammillary body damage (Kandel et al. 2021).

5. Anteroventral nucleus, left side. Part of the anterior thalamic nuclei; it relays the circuit to the cingulate cortex (Purves et al. 2018). *If it is damaged:* Damage to the anterior thalamus can cause amnesia (Kandel et

al. 2021).

6. Cingulum, left side. A long association bundle inside the cingulate gyrus that carries the circuit around the corpus callosum toward the parahippocampal gyrus (Papez 1937). *If it is damaged:* Surgical cuts of its front part (cingulotomy) have been used for intractable pain; they lessen the distress pain causes (Kandel et al. 2021).

7. Middle cingulate cortex and posterior cingulate cortex, left side. Papez placed the experience of emotion in the cingulate cortex; its middle and posterior parts pass the circuit on through the cingulum (Papez 1937). *If it is damaged:* Damage to the posterior cingulate cortex can impair memory and finding the way (Kandel et al. 2021).

8. Parahippocampal gyrus, left side. Contains the entorhinal cortex, the main gateway back into the hippocampus, which closes the loop (Purves et al. 2018). *If it is damaged:* The entorhinal cortex is among the first areas damaged in Alzheimer's disease, an early cause of memory loss (Braak and Braak 1991).

9. Hippocampus, left side. The circuit returns to the hippocampus through the perforant path from the entorhinal cortex (Purves et al. 2018). *If it is damaged:* As at the start: damage on both sides prevents new memories from forming (Scoville and Milner 1957).

What we simplified

We represent the anterior thalamus by its anteroventral nucleus alone, and the cingulate cortex by its middle and posterior parts, because our brain map divides the cingulate that way. The anterior cingulate is not part of this journey.

Question 15. Papez circuit

Is our account of this pathway accurate: the order of the stations, their sides, the role and the effect of a lesion at each, and the simplifications we made?

□ Correct as written
□ Incorrect
□ Correct after small changes
□ Outside my expertise or not sure

Voluntary movement (corticospinal)

It starts in the left motor cortex.

The corticospinal tract carries commands for voluntary movement from the motor cortex to the spinal cord. It runs down on its own side and crosses at the lower end of the medulla, so each hemisphere moves the opposite side of the body (Purves et al. 2018).

The journey

1. Precentral gyrus, left side. The primary motor cortex is mapped by body part; its large pyramidal neurons start the corticospinal tract (Penfield and Boldrey 1937). *If it is damaged:* Weakness of the right side of the body, mostly of the face and arm or of the leg depending on the part damaged, with upper motor neuron signs (Kandel et al. 2021).

2. Corticospinal tract, left side. Descends through the internal capsule, the cerebral peduncle, the pons and the medullary pyramid on the left side (Purves et al. 2018). *If it is damaged:* A lesion above the crossing, for example in the internal capsule, causes weakness of the right side of the body with spasticity, brisk reflexes and an upgoing plantar response (Kandel et al. 2021).

3. Crossing at the pyramidal decussation. At the junction of the medulla and the spinal cord most corticospinal fibres cross to the right side (Purves et al. 2018). *If it is damaged:* Above this point a lesion weakens the side of the body opposite to it, below this point the side of the lesion (Purves et al. 2018).

4. Lateral corticospinal tract, right side. Runs down the right lateral column of the spinal cord to the motor neurons of the right limbs (Purves et al. 2018). *If it is damaged:* Weakness below the lesion on the right side of the body, which is the side of the lesion, with upper motor neuron signs, as in Brown-Sequard syndrome (Kandel et al. 2021).

What we simplified

The final part of the journey, the lateral corticospinal tract in the spinal cord, is now shown in three dimensions from the PAM50 spinal cord atlas. The pyramidal decussation, where the fibres cross, is a step of its own described in words.

Question 16. Voluntary movement (corticospinal)

Is our account of this pathway accurate: the order of the stations, their sides, the role and the effect of a lesion at each, and the simplifications we made?

□ Correct as written
□ Incorrect
□ Correct after small changes
□ Outside my expertise or not sure

Touch and position (dorsal column, medial lemniscus)

It starts on the left side of the body.

The dorsal column and medial lemniscus pathway carries fine touch, vibration and position sense. It runs up the spinal cord on the side it enters, crosses in the medulla and reaches the thalamus and the somatosensory cortex of the opposite hemisphere (Purves et al. 2018).

The journey

1. Fasciculus gracilis and fasciculus cuneatus, left side. Fibres for fine touch, vibration and position sense run up the left dorsal column, gracile for the leg and cuneate for the arm, to their nuclei in the medulla (Purves et al. 2018). *If it is damaged:* Loss of fine touch, vibration and position sense on the left side below the lesion, with unsteadiness that worsens with the eyes closed (Kandel et al. 2021).

2. Crossing at the internal arcuate fibres. Axons of the gracile and cuneate nuclei cross the midline of the medulla and form the medial lemniscus (Purves et al. 2018). *If it is damaged:* From here on a lesion causes loss on the side of the body opposite to it (Purves et al. 2018).

3. Medial lemniscus, right side. Carries touch and position sense from the left side of the body up the right brainstem to the thalamus (Purves et al. 2018). *If it is damaged:* Loss of fine touch, vibration and position

sense on the left side of the body, for example in medial medullary syndrome (Kandel et al. 2021).

4. Ventral posterolateral nucleus, right side. The right ventral posterolateral nucleus is the thalamic relay for body sensation and projects to the primary somatosensory cortex (Purves et al. 2018). *If it is damaged:* Loss of sensation on the left side of the body, sometimes followed by severe pain on that side (thalamic pain syndrome) (Kandel et al. 2021).

5. Postcentral gyrus, right side. The primary somatosensory cortex, mapped by body part (Penfield and Boldrey 1937). *If it is damaged:* Loss of fine touch and position sense on the left side of the body, with difficulty recognising objects by touch (Kandel et al. 2021).

What we simplified

The dorsal columns of the spinal cord are now shown in three dimensions; the gracile and cuneate nuclei of the medulla, where they end, are described in words at the same station.

Question 17. Touch and position (dorsal column, medial lemniscus)

Is our account of this pathway accurate: the order of the stations, their sides, the role and the effect of a lesion at each, and the simplifications we made?

□ Correct as written
□ Incorrect
□ Correct after small changes
□ Outside my expertise or not sure

Pain and temperature (spinothalamic)

It starts on the left side of the body.

The spinothalamic pathway carries pain and temperature. Its fibres cross the spinal cord within a segment or two of where they enter, then run up the opposite side to the thalamus and the somatosensory cortex (Purves et al. 2018).

The journey

1. Dorsal horn, left side. Fibres for pain and temperature enter the spinal cord and end on neurons of the left dorsal horn near their level of entry (Purves et al. 2018). *If it is damaged:* Loss of pain and temperature in the area supplied by the damaged segments, on the left side (Kandel et al. 2021).

2. Crossing at the anterior white commissure. Axons of dorsal horn neurons cross in front of the central canal within one or two segments and form the spinothalamic tract (Purves et al. 2018). *If it is damaged:* A lesion around the central canal, for example syringomyelia, cuts the crossing fibres and causes loss of pain and temperature on both sides in a cape over the shoulders, with touch kept (Kandel et al. 2021).

3. Spinothalamic tract, right side. Carries pain and temperature from the left side of the body up the right side of the spinal cord and brainstem to the thalamus (Purves et al. 2018). *If it is damaged:* Loss of pain and temperature on the left side of the body, starting one or two segments below a spinal lesion; in Brown-Sequard syndrome this is the side opposite the weakness (Kandel et al. 2021).

4. Ventral posterolateral nucleus, right side. The right ventral posterolateral nucleus relays pain and temperature to the somatosensory cortex (Purves et al. 2018). *If it is damaged:* Loss of sensation on the left side

of the body, sometimes followed by severe burning pain there (thalamic pain syndrome) (Kandel et al. 2021).

5. Postcentral gyrus, right side. Places pain and temperature on the body map (Penfield and Boldrey 1937). *If it is damaged:* Poorer locating of pain and temperature on the left side of the body; the feeling of pain itself is often kept (Kandel et al. 2021).

What we simplified

The dorsal horn is now shown in three dimensions; the crossing in the anterior white commissure is a step of its own described in words.

Question 18. Pain and temperature (spinothalamic)

Is our account of this pathway accurate: the order of the stations, their sides, the role and the effect of a lesion at each, and the simplifications we made?

□ Correct as written
□ Incorrect
□ Correct after small changes
□ Outside my expertise or not sure

Basal ganglia motor loop

It starts in the left motor cortex.

The basal ganglia motor loop runs from the cortex through the striatum and the pallidum to the motor thalamus and back to the cortex, tuned by dopamine from the substantia nigra. It helps choose and start wanted movements and hold back unwanted ones, so its diseases cause too little movement, as in Parkinson's disease, or too much, as in Huntington's disease (Purves et al. 2018).

The journey

1. Supplementary motor area, left side. The supplementary motor area plans and starts self-initiated movements and sends this to the putamen (Purves et al. 2018). *If it is damaged:* Fewer self-started movements, mostly of the right limbs, and reduced speech after a left-sided lesion (Kandel et al. 2021).

2. Corticostriatal pathway, left side. Carries excitatory input from the cortex to the striatum (Purves et al. 2018). *If it is damaged:* Damage reduces the cortical input the basal ganglia need to select movements; it is rarely seen alone (Kandel et al. 2021).

3. Putamen, left side. The motor part of the striatum: it receives the cortex and starts the direct pathway, which releases movement, and the indirect pathway, which holds it back (Purves et al. 2018). *If it is damaged:* Can cause involuntary movements of the right limbs, such as dystonia or chorea; a putaminal bleed that reaches the internal capsule also weakens the right side (Kandel et al. 2021).

4. Substantia nigra, pars compacta, left side. Dopamine neurons of the pars compacta tune the putamen, helping the direct pathway and holding back the indirect one (Purves et al. 2018). *If it is damaged:* Loss of these neurons causes Parkinson's disease: slowness, rigidity and rest tremor, usually starting on the side of the body opposite the more affected nigra, here the right side (Kandel et al. 2021).

5. Globus pallidus, left side. The internal part of the globus pallidus is the main output of the basal ganglia; it keeps the motor thalamus under steady inhibition that the direct pathway releases (Purves et al. 2018). *If it*

is damaged: Lesions can cause dystonia; a surgical lesion (pallidotomy) eases Parkinson's symptoms on the right side (Kandel et al. 2021).

6. Ventral anterior nucleus and ventral lateral nucleus, left side. The motor thalamus relays the basal ganglia output back to the motor areas of the frontal cortex (Purves et al. 2018). *If it is damaged:* Damage can cause clumsiness and involuntary movements of the right limbs (Kandel et al. 2021).

7. Corticothalamic pathway, left side. Fibres between the motor thalamus and the frontal motor areas; in this loop they carry its output back to the cortex (Purves et al. 2018). *If it is damaged:* Damage, usually together with the internal capsule around it, causes weakness and clumsiness of the right limbs (Kandel et al. 2021).

8. Precentral gyrus, left side. The loop ends where movement commands leave the cortex, in the primary motor cortex (Penfield and Boldrey 1937). *If it is damaged:* Weakness of the right side of the body (Kandel et al. 2021).

What we simplified

We place the substantia nigra pars compacta as a station between the putamen and the globus pallidus, although it modulates the striatum from the side rather than lying in series. We also do not separate the direct and indirect pathways into their own stations; the text describes both at the putamen.

Question 19. Basal ganglia motor loop

Is our account of this pathway accurate: the order of the stations, their sides, the role and the effect of a lesion at each, and the simplifications we made?

□ Correct as written
□ Incorrect
□ Correct after small changes
□ Outside my expertise or not sure

Cerebellar motor loop

It starts in the left motor cortex.

The cerebellar loop runs from the motor cortex through the pons to the opposite cerebellar hemisphere, and back through the dentate nucleus, the red nucleus region and the thalamus to the motor cortex it started from. Because the loop crosses twice, each cerebellar hemisphere coordinates the limbs on its own side (Purves et al. 2018).

The journey

1. Precentral gyrus, left side. The motor cortex sends a copy of its commands to the cerebellum through the pons (Purves et al. 2018). *If it is damaged:* Weakness of the right side of the body (Kandel et al. 2021).

2. Frontopontine tract, left side. Carries cortical signals down the left cerebral peduncle to the pontine nuclei (Purves et al. 2018). *If it is damaged:* Rarely damaged alone; lesions of the cerebral peduncle also cut the corticospinal tract and weaken the right side (Kandel et al. 2021).

3. Pontine nuclei, left side. Relay neurons in the left half of the basal pons; their axons cross the midline (Purves et al. 2018). *If it is damaged:* Small pontine strokes can cause weakness with clumsiness of the right

limbs (ataxic hemiparesis) (Kandel et al. 2021).

4. Crossing through the middle cerebellar peduncle, in the midline. Pontine fibres cross the pons and enter the right cerebellar hemisphere through the middle cerebellar peduncle (Purves et al. 2018). *If it is damaged:* Clumsiness of the limbs on the side of the damaged peduncle (Kandel et al. 2021).

5. Cerebellar hemispheres, right side. The lateral cerebellum compares the planned movement with what is happening and times and coordinates it (Purves et al. 2018). *If it is damaged:* Clumsiness of the right limbs, the side of the lesion: poor aim with tremor near the target, trouble with rapid alternating movements and an unsteady gait (Kandel et al. 2021).

6. Dentate nucleus, right side. The deep nucleus of the lateral cerebellum; its output leaves through the superior cerebellar peduncle (Purves et al. 2018). *If it is damaged:* Clumsiness and intention tremor of the right limbs (Kandel et al. 2021).

7. Crossing through the superior cerebellar peduncle, in the midline. Dentate fibres leave through the superior cerebellar peduncle and cross in the midbrain to the left red nucleus and thalamus (Purves et al. 2018). *If it is damaged:* Damage below the crossing causes clumsiness on the side of the lesion, above it on the opposite side (Kandel et al. 2021).

8. Red nucleus, left side. The left red nucleus receives the crossed dentate fibres on their way to the thalamus (Purves et al. 2018). *If it is damaged:* Tremor and clumsiness of the right limbs, sometimes with an oculomotor palsy on the left side (Benedikt syndrome) (Kandel et al. 2021).

9. Ventral lateral nucleus, left side. The left ventral lateral nucleus relays the cerebellar output to the motor cortex (Purves et al. 2018). *If it is damaged:* Clumsiness of the right limbs; a surgical lesion here has been used to stop tremor (Kandel et al. 2021).

10. Precentral gyrus, left side. The loop closes: the motor cortex adjusts its commands with the cerebellar output (Purves et al. 2018). *If it is damaged:* Weakness of the right side of the body (Kandel et al. 2021).

What we simplified

We show the middle and superior cerebellar peduncles as crossing points in the midline, and route the dentate output through the red nucleus to the ventral lateral thalamus. Most dentatothalamic fibres pass by the red nucleus rather than relaying in it.

Question 20. Cerebellar motor loop

Is our account of this pathway accurate: the order of the stations, their sides, the role and the effect of a lesion at each, and the simplifications we made?

□ Correct as written
□ Incorrect
□ Correct after small changes
□ Outside my expertise or not sure

6. Behaviour circuits

Behaviour circuits show the brain systems behind reward, mood, arousal, attention, fear, stress, sleep, hunger, memory and decision making. They work on both sides of the brain at once, so there are no sides here. For every station we say what it does, which disorders are linked to it, and, where relevant, which medicines or substances act there.

Dopamine: reward and movement

It starts with the ventral tegmental area.

Dopamine neurons in the midbrain send three main projections: from the ventral tegmental area to the nucleus accumbens and limbic areas, to the prefrontal cortex, and from the substantia nigra to the dorsal striatum. Their bursts signal rewards that are better than expected, which drives learning and motivation (Schultz et al. 1997).

The journey

1. Ventral tegmental area. Dopamine neurons of the ventral tegmental area fire in bursts when a reward is better than predicted and pause when it is worse, a teaching signal for learning what to seek (Schultz et al. 1997). *Disorders:* Overactive signalling in this system is linked to the positive symptoms of schizophrenia and to the craving of addiction (Kandel et al. 2021). *Medicines and substances:* Opioids, alcohol and nicotine raise the firing of these neurons, part of why they are addictive (Kandel et al. 2021).

2. Medial forebrain bundle. The ascending dopamine axons travel forward through the lateral hypothalamus in the medial forebrain bundle (Purves et al. 2018). *Disorders:* Damage along this bundle can cause apathy and loss of motivation (Kandel et al. 2021).

3. Nucleus accumbens. The nucleus accumbens, the ventral striatum, turns reward signals into wanting and approach behaviour (Schultz et al. 1997). *Disorders:* Addiction changes this nucleus so that drug cues drive compulsive seeking (Kandel et al. 2021). *Medicines and substances:* Cocaine and amphetamine raise dopamine here by blocking or reversing its reuptake; antipsychotics block D2 receptors here (Kandel et al. 2021).

4. Superior frontal gyrus, medial part; superior frontal gyrus, medial orbital part; subgenual anterior cingulate cortex; and pregenual anterior cingulate cortex. Dopamine in the prefrontal and anterior cingulate cortex tunes working memory and goal-directed thinking (Miller and Cohen 2001). *Disorders:* Too little dopamine here is linked to the negative and cognitive symptoms of schizophrenia and to ADHD (Kandel et al. 2021). *Medicines and substances:* Methylphenidate raises dopamine and noradrenaline here, which helps attention in ADHD (Kandel et al. 2021).

5. Amygdala and hippocampus. Dopamine in the amygdala and hippocampus links places and cues with rewards and makes rewarding events easier to remember (Kandel et al. 2021). *Disorders:* Cue memories stored here keep the risk of relapse high long after drug use stops (Kandel et al. 2021).

6. Substantia nigra, pars compacta. Dopamine neurons of the substantia nigra pars compacta supply the dorsal striatum for starting and smoothing movement (Purves et al. 2018). *Disorders:* Their loss causes Parkinson's disease: slowness, rigidity and rest tremor, often with low mood (Kandel et al. 2021). *Medicines and substances:* L-DOPA is turned into dopamine by the remaining neurons; dopamine agonists act on the striatal receptors directly (Kandel et al. 2021).

7. Caudate nucleus and putamen. The caudate and putamen receive nigral dopamine, which favours the pathway that releases wanted movements and habits (Purves et al. 2018). *Disorders:* Blocking dopamine here causes parkinsonism and late involuntary movements (tardive dyskinesia) (Kandel et al. 2021). *Medicines and substances:* Antipsychotics that block D2 receptors here cause these movement side effects (Kandel et al. 2021).

What we simplified

Dopamine reaches far more targets than we show. We chose the nucleus accumbens, the prefrontal and anterior cingulate cortex, the amygdala and hippocampus, and the dorsal striatum as the stations a student most needs.

Question 21. Dopamine: reward and movement

Is our account of this circuit accurate: the stations, the role of each, the disorders and the medicines listed, and the simplifications we made?

□ Correct as written
□ Incorrect
□ Correct after small changes
□ Outside my expertise or not sure

Serotonin: mood and impulse

It starts with the raphe nuclei.

Serotonin comes from the raphe nuclei of the brainstem, whose axons reach almost the whole brain. It shapes mood, anxiety, impulse control, appetite and sleep, and most antidepressants act on it (Berger et al. 2009).

The journey

1. Dorsal raphe nucleus and median raphe nucleus. The dorsal and median raphe nuclei hold most of the serotonin neurons that project to the forebrain (Berger et al. 2009). *Disorders:* Low serotonin signalling is linked to depression, anxiety and impulsive aggression, although no single cause has been proven (Kandel et al. 2021). *Medicines and substances:* SSRIs block the serotonin transporter and raise serotonin around raphe targets; their effect on mood usually takes weeks (Kandel et al. 2021).

2. Hippocampus. Median raphe input to the hippocampus helps regulate memory and the response to stress (Berger et al. 2009). *Disorders:* In chronic stress and depression the hippocampus can shrink; antidepressant treatment may help it recover (Kandel et al. 2021).

3. Amygdala. Serotonin in the amygdala dampens or sharpens responses to threat (Berger et al. 2009). *Disorders:* Anxiety disorders are linked to an overactive amygdala that serotonin normally helps to restrain (Kandel et al. 2021). *Medicines and substances:* SSRIs are first-line medicines for many anxiety disorders (Kandel et al. 2021).

4. Subgenual anterior cingulate cortex; pregenual anterior cingulate cortex; and superior frontal gyrus, medial part. The subgenual and pregenual cingulate and medial prefrontal cortex, rich in serotonin receptors, help regulate mood (Berger et al. 2009). *Disorders:* The subgenual cingulate is overactive in depression (Kandel et al. 2021). *Medicines and substances:* Psilocybin and LSD act on serotonin 2A receptors in the cortex and change perception and mood (Berger et al. 2009).

5. Caudate nucleus, putamen and nucleus accumbens. Serotonin in the striatum and nucleus accumbens restrains impulsive and repetitive behaviour (Berger et al. 2009). *Disorders:* Obsessive-compulsive disorder is linked to these circuits (Kandel et al. 2021). *Medicines and substances:* High-dose SSRIs and clomipramine are used in obsessive-compulsive disorder (Kandel et al. 2021).

6. Suprachiasmatic nucleus. Raphe input to the suprachiasmatic nucleus links serotonin to the body clock and sleep (Saper et al. 2005). *Disorders:* Sleep is often disturbed in depression, with early waking (Kandel et al. 2021).

What we simplified

The raphe nuclei project to almost the whole brain. We show only the targets most relevant to mood and impulse control.

Question 22. Serotonin: mood and impulse

Is our account of this circuit accurate: the stations, the role of each, the disorders and the medicines listed, and the simplifications we made?

□ Correct as written
□ Incorrect
□ Correct after small changes
□ Outside my expertise or not sure

Noradrenaline: arousal and attention

It starts with the locus coeruleus.

Noradrenaline comes mainly from the locus coeruleus, a small nucleus in the pons whose branching axons reach nearly the whole brain. It sets the level of arousal and shifts attention between focused and broad modes (Aston-Jones and Cohen 2005).

The journey

1. Locus coeruleus. The locus coeruleus fires slowly in sleep, steadily when awake and in bursts when something important happens, adjusting how strongly the brain responds (Aston-Jones and Cohen 2005). *Disorders:* Overactivity is linked to anxiety, panic and the hyperarousal of PTSD; the nucleus is among the first sites damaged in Alzheimer's disease (Kandel et al. 2021). *Medicines and substances:* Clonidine reduces its firing and is used for withdrawal symptoms and ADHD (Kandel et al. 2021).

2. Dorsal noradrenergic bundle. Its axons run forward in the dorsal noradrenergic bundle to the thalamus and cortex (Purves et al. 2018). *Disorders:* Damage to ascending arousal fibres can lower alertness (Kandel et al. 2021).

3. Thalamus. Noradrenaline in the thalamus helps switch it from sleep rhythms to waking relay (Saper et al. 2005). *Disorders:* Weak arousal input to the thalamus contributes to drowsiness and reduced consciousness (Saper et al. 2005).

4. Frontal lobe and parietal lobe. In frontal and parietal cortex it sharpens attention and working memory (Aston-Jones and Cohen 2005). *Disorders:* ADHD is linked to weak prefrontal noradrenaline and dopamine signalling (Kandel et al. 2021). *Medicines and substances:* Atomoxetine and methylphenidate raise noradrenaline here and improve attention (Kandel et al. 2021).

5. Hippocampus and amygdala. In the hippocampus and amygdala it strengthens memories of emotional and stressful events (Kandel et al. 2021). *Disorders:* Overly strong fear memories in PTSD are linked to this effect (Kandel et al. 2021). *Medicines and substances:* Propranolol blocks beta receptors and is being studied as a way to weaken traumatic memories (Kandel et al. 2021).

6. Paraventricular nucleus. It drives the paraventricular nucleus to start the hormonal stress response (Ulrich-Lai and Herman 2009). *Disorders:* Chronic stress keeps this drive high (Ulrich-Lai and Herman 2009).

7. Cerebellar hemispheres. Projections to the cerebellum and spinal cord help adjust movement to the level of arousal (Purves et al. 2018). *Disorders:* Stress and anxiety increase physiological tremor through noradrenaline (Kandel et al. 2021). *Medicines and substances:* Propranolol reduces stress-related tremor and the bodily symptoms of performance anxiety (Kandel et al. 2021).

What we simplified

We show the thalamus, the frontal and parietal cortex, the hippocampus and amygdala, the paraventricular nucleus and the cerebellum as targets of the locus coeruleus.

Question 23. Noradrenaline: arousal and attention

Is our account of this circuit accurate: the stations, the role of each, the disorders and the medicines listed, and the simplifications we made?

□ Correct as written
□ Incorrect
□ Correct after small changes
□ Outside my expertise or not sure

Acetylcholine: attention and memory

It starts with the basal forebrain.

Acetylcholine in the brain comes from the basal forebrain, which supplies the cortex and hippocampus, and from the brainstem, which supplies the thalamus. It raises attention to new information and helps store new memories (Hasselmo 2006).

The journey

1. Nucleus basalis of Meynert. The nucleus basalis of Meynert is the main source of acetylcholine for the cortex (Hasselmo 2006). *Disorders:* It degenerates early in Alzheimer's disease, and its loss contributes to the memory decline (Kandel et al. 2021). *Medicines and substances:* Donepezil and other cholinesterase in-

hibitors slow the breakdown of acetylcholine and modestly help memory in Alzheimer's disease (Kandel et al. 2021).

2. Cerebral cortex. Across the cortex acetylcholine strengthens incoming sensory signals over internal activity, which supports attention (Hasselmo 2006). *Disorders:* Loss of cortical acetylcholine contributes to inattention and confusion in dementia (Kandel et al. 2021). *Medicines and substances:* Nicotine acts on cortical nicotinic receptors and briefly sharpens attention (Kandel et al. 2021).

3. Medial septal nucleus. The medial septal nucleus supplies the hippocampus with acetylcholine and paces its theta rhythm (Hasselmo 2006). *Disorders:* Damage to the septum impairs memory for places (Kandel et al. 2021).

4. Hippocampus. In the hippocampus acetylcholine favours storing new memories over recalling old ones (Hasselmo 2006). *Disorders:* Low hippocampal acetylcholine contributes to forgetfulness in Alzheimer's disease (Kandel et al. 2021). *Medicines and substances:* Anticholinergic medicines such as scopolamine impair new memories and can cause delirium in older people (Kandel et al. 2021).

5. Pedunculopontine nucleus and laterodorsal tegmental nucleus. The pedunculopontine and laterodorsal tegmental nuclei of the brainstem release acetylcholine onto the thalamus during waking and dream sleep (Saper et al. 2005). *Disorders:* Their loss in Parkinson's disease is linked to falls and gait problems (Kandel et al. 2021).

6. Thalamus. In the thalamus acetylcholine helps open the relay to the cortex during wakefulness and REM sleep (Saper et al. 2005). *Disorders:* Anticholinergic overload disturbs this relay and contributes to delirium (Kandel et al. 2021).

What we simplified

The medial septal nucleus is described in words only, because the atlas does not yet show it. The brainstem cholinergic nuclei, the pedunculopontine and laterodorsal tegmental nuclei, are now shown.

Question 24. Acetylcholine: attention and memory

Is our account of this circuit accurate: the stations, the role of each, the disorders and the medicines listed, and the simplifications we made?

□ Correct as written
□ Incorrect
□ Correct after small changes
□ Outside my expertise or not sure

Fear circuit

It starts with the sound of danger.

The fear circuit centres on the amygdala. A fast route from the sensory thalamus and a slower one through the cortex bring signals of danger; the amygdala then drives freezing, stress hormones and arousal, while the hippocampus adds context and the prefrontal cortex calms the response (LeDoux 2000).

The journey

1. Medial geniculate nucleus. The medial geniculate nucleus sends a fast, rough copy of a sound straight to the amygdala, before the cortex has analysed it (LeDoux 2000). *Disorders:* This quick route lets learned danger cues trigger fear before they are consciously recognised (LeDoux 2000).

2. Amygdala. The amygdala learns which cues predict danger; its central nucleus sends the commands for the fear response (LeDoux 2000). *Disorders:* An overactive amygdala is linked to anxiety disorders and PTSD; damage to both amygdalae reduces fear (Kandel et al. 2021). *Medicines and substances:* Benzodiazepines enhance GABA inhibition in the amygdala and reduce anxiety (Kandel et al. 2021).

3. Lateral hypothalamus and paraventricular nucleus. The lateral hypothalamus raises heart rate and blood pressure through the autonomic system, and the paraventricular nucleus starts the stress hormones (LeDoux 2000). *Disorders:* Repeated activation contributes to the bodily symptoms of chronic anxiety (Kandel et al. 2021).

4. Periaqueductal grey. The periaqueductal grey in the midbrain organises freezing and, under greater threat, fight or flight (LeDoux 2000). *Disorders:* Its activation is thought to underlie the intense bodily feelings of panic attacks (Kandel et al. 2021).

5. Locus coeruleus. The locus coeruleus raises vigilance and arousal during fear (Aston-Jones and Cohen 2005). *Disorders:* Persistent hyperarousal in PTSD is linked to this system (Kandel et al. 2021). *Medicines and substances:* Prazosin, which blocks noradrenaline alpha-1 receptors, is used for PTSD nightmares (Kandel et al. 2021).

6. Hippocampus. The hippocampus tells the amygdala where and when danger occurred, so fear fits the context (LeDoux 2000). *Disorders:* Weak context signalling may let fear spread to safe situations in PTSD (Kandel et al. 2021).

7. Subgenual anterior cingulate cortex; and superior frontal gyrus, medial part. The medial prefrontal and subgenual cingulate cortex restrain the amygdala when a cue is no longer dangerous (extinction) (LeDoux 2000). *Disorders:* Weak prefrontal control of the amygdala is linked to persistent anxiety and PTSD (Kandel et al. 2021). *Medicines and substances:* Exposure therapy relies on this extinction learning; SSRIs support it in PTSD (Kandel et al. 2021).

What we simplified

We show the fast route from the thalamus to the amygdala as its own station; the slower route through the cortex is mentioned only in the overview. The periaqueductal grey is now shown as its own station.

Question 25. Fear circuit

Is our account of this circuit accurate: the stations, the role of each, the disorders and the medicines listed, and the simplifications we made?

□ Correct as written
□ Incorrect
□ Correct after small changes
□ Outside my expertise or not sure

Stress axis (HPA)

It starts with a stressor.

The stress axis (hypothalamic-pituitary-adrenal axis) turns a perceived threat into a rise of cortisol within minutes. Cortisol mobilises energy and, through feedback on the brain and pituitary, switches the response off again (Ulrich-Lai and Herman 2009).

The journey

1. Amygdala. The amygdala drives the axis during threat and becomes more excitable under cortisol (Ulrich-Lai and Herman 2009). *Disorders:* In PTSD the axis is often set unusually, with altered cortisol levels and strong feedback (Kandel et al. 2021).

2. Paraventricular nucleus. Neurons of the paraventricular nucleus release corticotropin-releasing hormone (CRH) into the pituitary portal blood (Ulrich-Lai and Herman 2009). *Disorders:* Overactive CRH release is found in many people with severe depression (Kandel et al. 2021).

3. Pituitary gland. The anterior pituitary answers CRH by releasing adrenocorticotropic hormone (ACTH) into the blood within minutes (Ulrich-Lai and Herman 2009). *Disorders:* An ACTH-secreting pituitary adenoma causes Cushing's disease (Kandel et al. 2021).

4. Adrenal cortex. ACTH makes the adrenal cortex release cortisol, which peaks about 20 to 40 minutes after a short stressor (Dickerson and Kemeny 2004). *Disorders:* Too much cortisol causes Cushing's syndrome with weight gain, high blood sugar and low mood; too little (Addison's disease) causes weakness and low blood pressure (Kandel et al. 2021). *Medicines and substances:* Glucocorticoid medicines such as prednisolone act like cortisol and, taken for long, suppress the axis (Kandel et al. 2021).

5. Hippocampus. Cortisol acts on hippocampal receptors, which signal the hypothalamus to end the stress response (Ulrich-Lai and Herman 2009). *Disorders:* Long-term high cortisol may damage the hippocampus and weaken this brake, as reported in chronic stress and depression (Kandel et al. 2021).

6. Superior frontal gyrus, medial part; and subgenual anterior cingulate cortex. The medial prefrontal and subgenual cingulate cortex also inhibit the axis and help recovery after stress (Ulrich-Lai and Herman 2009). *Disorders:* Weak prefrontal control is linked to prolonged stress responses in depression (Kandel et al. 2021).

What we simplified

The journey begins at the amygdala, where a threat is recognised, and then follows the hypothalamus, the pituitary and the adrenal cortex, ending with the feedback from the hippocampus and the prefrontal cortex. The adrenal cortex lies outside the brain and is described in words only.

Question 26. Stress axis (HPA)

Is our account of this circuit accurate: the stations, the role of each, the disorders and the medicines listed, and the simplifications we made?

□ Correct as written
□ Incorrect
□ Correct after small changes
□ Outside my expertise or not sure

Body clock, sleep and waking

It starts with light on the retina.

The suprachiasmatic nucleus is the master clock: its neurons keep a roughly 24-hour rhythm and are reset each day by light. It times melatonin at night and the morning cortisol rise, and works with arousal centres and a sleep switch in the hypothalamus (Saper et al. 2005).

The journey

1. Retinohypothalamic tract. Light-sensitive retinal ganglion cells send light signals to the suprachiasmatic nucleus through the retinohypothalamic tract (Saper et al. 2005). *Disorders:* Blindness without any light perception can leave the clock running out of step with the day (Kandel et al. 2021).

2. Suprachiasmatic nucleus. The suprachiasmatic nucleus keeps the 24-hour rhythm and resets it from the light signal (Saper et al. 2005). *Disorders:* Jet lag and shift work put the clock out of step with the day; delayed sleep phase shifts it later (Kandel et al. 2021). *Medicines and substances:* Melatonin taken in the evening can shift the clock earlier (Saper et al. 2005).

3. Paraventricular nucleus. The paraventricular nucleus passes the clock signal to the sympathetic pathway that controls the pineal gland, and times the morning cortisol rise (Saper et al. 2005). *Disorders:* A flattened cortisol rhythm is common in depression and chronic stress (Kandel et al. 2021).

4. Pineal gland. The pineal gland releases melatonin at night, signalling darkness to the body (Saper et al. 2005). *Disorders:* Bright light at night suppresses melatonin and can delay sleep (Kandel et al. 2021). *Medicines and substances:* Melatonin and melatonin agonists help some sleep-onset and jet-lag problems (Kandel et al. 2021).

5. Lateral hypothalamus. Orexin (hypocretin) neurons of the lateral hypothalamus keep wakefulness stable (Saper et al. 2005). *Disorders:* Their loss causes narcolepsy, with sleep attacks and cataplexy (Thannickal et al. 2000). *Medicines and substances:* Orexin receptor blockers such as suvorexant are used for insomnia (Kandel et al. 2021).

6. Tuberomammillary nucleus, locus coeruleus, dorsal raphe nucleus and median raphe nucleus. The tuberomammillary nucleus (histamine), locus coeruleus (noradrenaline) and raphe (serotonin) form the arousal system that keeps the cortex awake (Saper et al. 2005). *Disorders:* Weak arousal signalling contributes to daytime sleepiness (Saper et al. 2005). *Medicines and substances:* Older antihistamines cause drowsiness by blocking histamine; modafinil and caffeine promote wakefulness (Kandel et al. 2021).

7. Thalamus and cerebral cortex. The thalamus and cortex switch from slow sleep rhythms to waking activity under this arousal input (Saper et al. 2005). *Disorders:* Damage to the arousal input of the thalamus can cause long-lasting loss of consciousness (Kandel et al. 2021).

8. Ventrolateral preoptic nucleus. The ventrolateral preoptic nucleus inhibits the arousal centres to switch on sleep, and they in turn inhibit it, forming a flip-flop switch (Saper et al. 2005). *Disorders:* Its loss with age is linked to broken sleep (Saper et al. 2005). *Medicines and substances:* Benzodiazepine-type sleeping pills enhance GABA inhibition, the signal this nucleus uses (Kandel et al. 2021).

What we simplified

The tuberomammillary nucleus, the locus coeruleus and the raphe nuclei are grouped into one station as the arousal system. The sleep switch in the ventrolateral preoptic nucleus is described in words only.

Question 27. Body clock, sleep and waking

Is our account of this circuit accurate: the stations, the role of each, the disorders and the medicines listed, and the simplifications we made?

□ Correct as written
□ Incorrect
□ Correct after small changes
□ Outside my expertise or not sure

Hunger and satiety

It starts with gut and fat signals.

Hunger and fullness are set mainly in the hypothalamus, which reads hormones from the gut and fat, and in the brainstem, which reads signals from the stomach. Reward and taste areas decide how much we want a particular food (Morton et al. 2006).

The journey

1. Ghrelin, leptin and insulin. Ghrelin from the empty stomach rises before meals and signals hunger; leptin from fat and insulin from the pancreas signal stored and incoming energy (Cummings et al. 2001). *Disorders:* Lack of leptin causes severe obesity from childhood; most obesity involves leptin resistance instead (Morton et al. 2006).

2. Arcuate nucleus. The arcuate nucleus holds two opposing groups: AgRP and NPY neurons that drive hunger and POMC neurons that signal fullness (Morton et al. 2006). *Disorders:* Faults in this melanocortin pathway, such as MC4R mutations, are the commonest single-gene cause of obesity (Morton et al. 2006). *Medicines and substances:* GLP-1 medicines such as semaglutide act on the arcuate nucleus and brainstem to reduce appetite (Kandel et al. 2021).

3. Paraventricular nucleus. The paraventricular nucleus receives the arcuate signals and turns them into fullness and energy use (Morton et al. 2006). *Disorders:* Damage to this area, for example by tumours near the hypothalamus, can cause uncontrolled eating and obesity (Morton et al. 2006).

4. Lateral hypothalamus. The lateral hypothalamus drives hunger and food seeking, partly through orexin and MCH neurons (Morton et al. 2006). *Disorders:* Lesions of the lateral hypothalamus cause loss of appetite and weight in animals (Morton et al. 2006). *Medicines and substances:* Cannabinoids increase appetite partly through receptors in the hypothalamus (Morton et al. 2006).

5. Ventromedial hypothalamic nucleus. The ventromedial nucleus contributes to fullness and to the body's response to low blood sugar (Morton et al. 2006). *Disorders:* Damage to the ventromedial hypothalamus causes overeating and obesity (hypothalamic obesity) (Kandel et al. 2021).

6. Viscero-sensory-motor nuclei (solitary tract and dorsal vagal nuclei). The nucleus of the solitary tract in the brainstem receives vagal signals of stomach stretch and gut hormones and helps end a meal (Morton et al. 2006). *Disorders:* Disturbed gut-brain signals are thought to contribute to eating disorders (Kandel et al.

2021). *Medicines and substances:* GLP-1 medicines also act here and slow the emptying of the stomach (Kandel et al. 2021).

7. Ventral tegmental area and nucleus accumbens. The ventral tegmental area and nucleus accumbens make tasty food rewarding and drive eating beyond need (Morton et al. 2006). *Disorders:* Food reward is thought to be altered in binge eating and in anorexia nervosa (Kandel et al. 2021).

8. Insula, medial orbital gyrus, anterior orbital gyrus, posterior orbital gyrus and lateral orbital gyrus. The insula holds the primary taste cortex, and the orbitofrontal cortex values a food, a value that falls as we eat it (Kandel et al. 2021). *Disorders:* Altered insula and orbitofrontal responses to food are reported in anorexia nervosa (Kandel et al. 2021).

What we simplified

The hormones from the gut and fat tissue are described in words only. The nucleus of the solitary tract in the brainstem is now shown, together with the dorsal vagal nucleus beside it.

Question 28. Hunger and satiety

Is our account of this circuit accurate: the stations, the role of each, the disorders and the medicines listed, and the simplifications we made?

□ Correct as written
□ Incorrect
□ Correct after small changes
□ Outside my expertise or not sure

Memory

It starts with an experience.

The brain keeps several kinds of memory. Facts and events depend on the hippocampus and nearby cortex while they are being stored and are later held in the cortex; habits and skills depend on the striatum and cerebellum (Squire 2004).

The journey

1. Fusiform gyrus, middle temporal gyrus and inferior temporal gyrus. Temporal association cortex processes what we perceive and later holds the long-term traces of facts and events (Squire 2004). *Disorders:* Damage to the lateral temporal cortex causes loss of knowledge about words and objects (semantic dementia) (Kandel et al. 2021).

2. Parahippocampal gyrus. The parahippocampal gyrus and entorhinal cortex gather information from the cortex and pass it to the hippocampus (Squire 2004). *Disorders:* The entorhinal cortex is among the first areas damaged in Alzheimer's disease (Braak and Braak 1991).

3. Hippocampus. The hippocampus binds the parts of an experience into a new memory (Squire 2004). *Disorders:* Patient HM could no longer form new memories after both hippocampi were removed, while old memories and skills remained (Scoville and Milner 1957). *Medicines and substances:* Benzodiazepines and alcohol in high doses block the storage of new memories (blackouts) (Kandel et al. 2021).

4. Fornix and mammillary body. The fornix carries hippocampal output to the mammillary bodies, part of the circuit needed for recall (Squire 2004). *Disorders:* Thiamine deficiency damages the mammillary bodies and causes Korsakoff amnesia (Kandel et al. 2021). *Medicines and substances:* Thiamine given early prevents and treats Wernicke-Korsakoff syndrome (Kandel et al. 2021).

5. Superior frontal gyrus, medial part. Over time memories are reorganised so that the medial prefrontal cortex and other cortex can retrieve them with less help from the hippocampus (Squire 2004). *Disorders:* Frontal damage causes problems with recalling the source and order of events (Kandel et al. 2021).

6. Amygdala. The amygdala strengthens memories of emotional events, which is why they are vivid (Kandel et al. 2021). *Disorders:* Overly strong emotional memories contribute to PTSD (Kandel et al. 2021).

7. Nucleus basalis of Meynert. Acetylcholine from the nucleus basalis supports attention and the storage of new memories (Hasselmo 2006). *Disorders:* Its loss contributes to the memory problems of Alzheimer's disease (Kandel et al. 2021). *Medicines and substances:* Cholinesterase inhibitors such as donepezil are used to support memory in Alzheimer's disease (Kandel et al. 2021).

8. Caudate nucleus, putamen and cerebellar hemispheres. The striatum and cerebellum store habits and motor skills, which do not need the hippocampus (Squire 2004). *Disorders:* Parkinson's disease impairs habit learning, while amnesic patients like HM can still learn new skills (Squire 2004).

What we simplified

We put the hippocampal system for facts and events and the striatal and cerebellar systems for habits and skills on one journey, to let students contrast them.

Question 29. Memory

Is our account of this circuit accurate: the stations, the role of each, the disorders and the medicines listed, and the simplifications we made?

□ Correct as written
□ Incorrect
□ Correct after small changes
□ Outside my expertise or not sure

Decision and self-control

It starts with a choice.

Deciding and controlling oneself depend on the prefrontal cortex, which holds goals in mind and biases other areas toward them, working in loops with the striatum and thalamus (Miller and Cohen 2001).

The journey

1. Superior frontal gyrus, dorsolateral part; and middle frontal gyrus. The dorsolateral prefrontal cortex holds goals and rules in working memory and plans steps toward them (Miller and Cohen 2001). *Disorders:* Damage impairs planning and flexible thinking; reduced function is reported in schizophrenia (Kandel et al. 2021). *Medicines and substances:* Methylphenidate and atomoxetine improve prefrontal control in ADHD (Kandel et al. 2021).

2. Medial orbital gyrus, anterior orbital gyrus, posterior orbital gyrus and lateral orbital gyrus. The orbitofrontal cortex judges the value of options and updates it when outcomes change (Kandel et al. 2021). *Disorders:* Overactivity in orbitofrontal-striatal loops is linked to obsessive-compulsive disorder (Kandel et al. 2021). *Medicines and substances:* High-dose SSRIs reduce the symptoms of obsessive-compulsive disorder (Kandel et al. 2021).

3. Superior frontal gyrus, medial part; and superior frontal gyrus, medial orbital part. The ventromedial prefrontal cortex links decisions to emotions and social rules (Damasio et al. 1994). *Disorders:* Phineas Gage became impulsive and socially careless after an iron bar damaged this region in 1848 (Damasio et al. 1994).

4. Pregenual anterior cingulate cortex and supracallosal anterior cingulate cortex. The anterior cingulate cortex detects conflicts and errors and calls for more control (Miller and Cohen 2001). *Disorders:* Altered anterior cingulate function is reported in ADHD and obsessive-compulsive disorder (Kandel et al. 2021).

5. Inferior frontal gyrus, opercular part; and inferior frontal gyrus, triangular part. The inferior frontal gyrus, especially on the right, helps stop actions that are no longer wanted (Kandel et al. 2021). *Disorders:* Weak stopping is linked to impulsivity in ADHD and addiction (Kandel et al. 2021).

6. Mediodorsal nucleus, medial magnocellular part; and mediodorsal nucleus, lateral parvocellular part. The mediodorsal nucleus of the thalamus links the prefrontal cortex with the basal ganglia loop and keeps goal information active (Miller and Cohen 2001). *Disorders:* Damage here can cause apathy and memory problems similar to frontal damage (Kandel et al. 2021).

7. Caudate nucleus and nucleus accumbens. The caudate and nucleus accumbens weigh effort against reward and turn choices into action (Schultz et al. 1997). *Disorders:* Addiction biases these loops toward drug rewards over long-term goals (Kandel et al. 2021).

What we simplified

The stations follow the order in which decision making is usually taught, not a strict anatomical sequence.

Question 30. Decision and self-control

Is our account of this circuit accurate: the stations, the role of each, the disorders and the medicines listed, and the simplifications we made?

□ Correct as written
□ Incorrect
□ Correct after small changes
□ Outside my expertise or not sure

7. Hormone rhythms

For three circuits the atlas plays a simple picture of how hormone levels change over time while the journey moves from station to station. These pictures are schematic: each curve is scaled to its own peak and shows a shape, not a laboratory value. We describe each picture in words below.

The stress response

The picture covers two hours after a short stressor at time zero. Corticotropin-releasing hormone rises within the first two minutes and peaks at about five minutes. Adrenocorticotropic hormone follows and peaks at about ten minutes. Cortisol rises from about ten minutes, peaks at about thirty minutes, and has almost returned to its starting level by two hours.

As the picture plays, the journey moves from the amygdala at the start, to the paraventricular nucleus at two minutes, the pituitary at eight, the adrenal cortex at twenty-five, the hippocampal feedback at fifty and the prefrontal cortex at seventy minutes.

Schematic levels after a short stressor at time 0, each scaled to its own peak: CRH acts within minutes, ACTH peaks after about 10 minutes and cortisol after 20 to 40 minutes, then feedback returns them to baseline (Dickerson and Kemeny 2004) (Ulrich-Lai and Herman 2009).

Question 31. The stress response

Are these shapes and timings right for teaching?

□ Correct as written
□ Incorrect
□ Correct after small changes
□ Outside my expertise or not sure

The daily cycle

The picture covers one day from midnight in an adult who is awake by day. Melatonin is high through the night, peaks at about three in the morning, falls to a low level by eight, stays low through the day and rises again from about nine in the evening. Cortisol is lowest around midnight, starts to rise at about three in the morning, peaks at about eight, shortly after waking, and declines through the day.

The journey shows the pineal gland at night, the light signal and the master clock at dawn, the morning cortisol signal at seven, the wakefulness centres through the morning, and the sleep switch at ten in the evening.

Schematic 24-hour levels from midnight in a day-active adult, each scaled to its own peak: melatonin is high at night and cortisol peaks soon after waking (Weitzman et al. 1971) (Saper et al. 2005).

Question 32. The daily cycle

Are these shapes and timings right for teaching?

□ Correct as written
□ Incorrect
□ Correct after small changes
□ Outside my expertise or not sure

The meal cycle

The picture covers one day with meals at seven, thirteen and nineteen hours. Ghrelin rises in the hour before each meal and falls within an hour after it; it is moderately high through the night. Insulin is low between meals, rises sharply with each meal and returns towards its low level within about two hours.

The journey follows the first meal of the day in detail: the hunger signals in the arcuate nucleus and lateral hypothalamus before breakfast, the reward and taste stations while eating, the brainstem signal that ends the meal, and the fullness signals of the paraventricular and ventromedial nuclei afterwards.

Schematic levels over a day with meals at 7, 13 and 19 h, each scaled to its own peak: ghrelin rises before each meal and falls after it, while insulin rises with each meal (Cummings et al. 2001) (Morton et al. 2006).

Question 33. The meal cycle

Are these shapes and timings right for teaching?

□ Correct as written
□ Incorrect
□ Correct after small changes
□ Outside my expertise or not sure

8. The statements we are least sure of

These fifteen statements were the hardest to support while writing. Each also appears in its journey above; we list them again because we would value your opinion on them most. For each we give the statement, the reason for our doubt and the source we relied on.

The mammillothalamic tract and memory

What we wrote: Thalamic strokes that cut the mammillothalamic tract can cause memory loss like that of mammillary body damage. (Kandel et al. 2021)

Why we hesitate: Isolated damage to the tract is rare. Most reported cases also involve thalamic nuclei, so it is hard to know how much of the amnesia the tract alone explains.

Question 34. The mammillothalamic tract and memory

Is this statement accurate?

□ Correct as written
□ Incorrect
□ Correct after small changes
□ Outside my expertise or not sure

Cingulotomy and pain

What we wrote: Surgical cutting of the front part of the cingulum has been used for intractable pain and lessens the distress that pain causes. (Kandel et al. 2021)

Why we hesitate: The procedure is rare and much of the evidence comes from older, small case series. We are not sure the statement reflects current understanding.

Question 35. Cingulotomy and pain

Is this statement accurate?

□ Correct as written
□ Incorrect
□ Correct after small changes
□ Outside my expertise or not sure

The posterior cingulate cortex

What we wrote: Damage to the posterior cingulate cortex can impair memory and finding the way. (Kandel et al. 2021)

Why we hesitate: Much of the evidence concerns the neighbouring retrosplenial cortex, and the boundary between the two is not always drawn the same way.

Question 36. The posterior cingulate cortex

Is this statement accurate?

□ Correct as written
□ Incorrect
□ Correct after small changes
□ Outside my expertise or not sure

The supplementary motor area and speech

What we wrote: A lesion of the supplementary motor area causes fewer self-started movements of the opposite limbs and reduced speech after a left-sided lesion. (Kandel et al. 2021)

Why we hesitate: These effects are often temporary and their link to the side of the lesion varies between reports.

Question 37. The supplementary motor area and speech

Is this statement accurate?

□ Correct as written
□ Incorrect
□ Correct after small changes
□ Outside my expertise or not sure

Damage to the corticostriatal pathway

What we wrote: Damage to the corticostriatal pathway reduces the cortical input the basal ganglia need to select movements, and is rarely seen alone. (Kandel et al. 2021)

Why we hesitate: This statement is vague, and we are not sure it teaches anything useful. It may be better replaced.

Question 38. Damage to the corticostriatal pathway

Is this statement accurate?

□ Correct as written
□ Incorrect
□ Correct after small changes
□ Outside my expertise or not sure

The trapezoid body

What we wrote: Damage at the trapezoid body mainly impairs locating sounds, because the superior olivary complex compares the two ears. (Purves et al. 2018)

Why we hesitate: Isolated human lesions here are rare; the reasoning comes mostly from physiology and animal studies.

Question 39. The trapezoid body

Is this statement accurate?

□ Correct as written
□ Incorrect
□ Correct after small changes
□ Outside my expertise or not sure

The inferior colliculus

What we wrote: A one-sided lesion of the inferior colliculus causes mostly mild difficulty locating sounds, while lesions on both sides can cause marked hearing loss. (Kandel et al. 2021)

Why we hesitate: Human cases are few and varied.

Question 40. The inferior colliculus

Is this statement accurate?

□ Correct as written
□ Incorrect
□ Correct after small changes
□ Outside my expertise or not sure

The acoustic radiation

What we wrote: A one-sided lesion of the acoustic radiation does not cause deafness but impairs locating sounds coming from the opposite side of space. (Kandel et al. 2021)

Why we hesitate: Deficits in sound localisation after one-sided lesions vary between patients, and we are not certain the side is stated correctly.

Question 41. The acoustic radiation

Is this statement accurate?

□ Correct as written
□ Incorrect
□ Correct after small changes
□ Outside my expertise or not sure

Pontine strokes

What we wrote: Small pontine strokes can cause weakness with clumsiness of the opposite limbs (ataxic hemiparesis). (Kandel et al. 2021)

Why we hesitate: Ataxic hemiparesis also follows lesions of the internal capsule and thalamus, so presenting it under the pontine nuclei may mislead.

Question 42. Pontine strokes

Is this statement accurate?

□ Correct as written
□ Incorrect
□ Correct after small changes
□ Outside my expertise or not sure

The medial forebrain bundle

What we wrote: Damage along the medial forebrain bundle can cause apathy and loss of motivation. (Kandel et al. 2021)

Why we hesitate: Human evidence is limited and much of it comes from animal studies.

Question 43. The medial forebrain bundle

Is this statement accurate?

□ Correct as written
□ Incorrect
□ Correct after small changes
□ Outside my expertise or not sure

Noradrenaline and tremor

What we wrote: Stress and anxiety increase physiological tremor through noradrenaline, and propranolol reduces this tremor. (Kandel et al. 2021)

Why we hesitate: The tremor is driven largely by beta receptors outside the brain, on the muscles themselves, so attaching it to the brain's noradrenergic projections may be misleading.

Question 44. Noradrenaline and tremor

Is this statement accurate?

□ Correct as written
□ Incorrect
□ Correct after small changes
□ Outside my expertise or not sure

The septum and spatial memory

What we wrote: Damage to the septum impairs memory for places. (Kandel et al. 2021)

Why we hesitate: The evidence is mainly from animals; human septal lesions are rare and usually come with damage to neighbouring structures.

Question 45. The septum and spatial memory

Is this statement accurate?

□ Correct as written
□ Incorrect
□ Correct after small changes
□ Outside my expertise or not sure

The periaqueductal grey and panic

What we wrote: Activation of the periaqueductal grey is thought to underlie the intense bodily feelings of panic attacks. (Kandel et al. 2021)

Why we hesitate: This is a theory based on animal and stimulation studies and is not established in people.

Question 46. The periaqueductal grey and panic

Is this statement accurate?

□ Correct as written
□ Incorrect
□ Correct after small changes
□ Outside my expertise or not sure

The paraventricular nucleus and overeating

What we wrote: Damage to the paraventricular area, for example by tumours near the hypothalamus, can cause uncontrolled eating and obesity. (Morton et al. 2006)

Why we hesitate: Hypothalamic obesity after tumours is usually linked to damage of the medial and ventromedial hypothalamus as a whole, not specifically to the paraventricular nucleus.

Question 47. The paraventricular nucleus and overeating

Is this statement accurate?

□ Correct as written
□ Incorrect
□ Correct after small changes
□ Outside my expertise or not sure

The mediodorsal thalamus

What we wrote: Damage to the mediodorsal nucleus of the thalamus can cause apathy and memory problems similar to frontal damage. (Kandel et al. 2021)

Why we hesitate: Thalamic amnesia is more often linked to the anterior nuclei and the mammillothalamic tract; the role of the mediodorsal nucleus is debated.

Question 48. The mediodorsal thalamus

Is this statement accurate?

□ Correct as written
□ Incorrect
□ Correct after small changes
□ Outside my expertise or not sure

9. Our sources

Every statement in this booklet names its source. Most clinical statements cite two standard textbooks as a whole, Purves and colleagues' Neuroscience and Kandel and colleagues' Principles of Neural Science, without a chapter or page. Several statements about behaviour and hormones cite review articles and classic research papers. The full list follows the two questions below.

Question 49. Textbooks cited as a whole

Are these two textbooks appropriate sources for the clinical statements? If you can, please suggest chapters or better sources in the notes.

□ Correct as written
□ Correct after small changes
□ Incorrect
□ Outside my expertise or not sure

Question 50. Research papers

Are the research papers in the list appropriate support for the statements that cite them? If a better or more current source exists, please name it in the notes.

□ Correct as written
□ Correct after small changes
□ Incorrect
□ Outside my expertise or not sure

The full list

1. Aston-Jones G, Cohen JD (2005). An integrative theory of locus coeruleus-norepinephrine function: adaptive gain and optimal performance. Annual Review of Neuroscience 28: 403-450. doi:10.1146/annurev.neuro.28.061604.135709 https://doi.org/10.1146/annurev.neuro.28.061604.135709
2. Berger M, Gray JA, Roth BL (2009). The expanded biology of serotonin. Annual Review of Medicine 60: 355-366. doi:10.1146/annurev.med.60.042307.110802 https://doi.org/10.1146/annurev.med.60.042307.110802
3. Braak H, Braak E (1991). Neuropathological stageing of Alzheimer-related changes. Acta Neuropathologica 82(4): 239-259. doi:10.1007/BF00308809 https://doi.org/10.1007/BF00308809
4. Cummings DE, Purnell JQ, Frayo RS, Schmidova K, Wisse BE, Weigle DS (2001). A preprandial rise in plasma ghrelin levels suggests a role in meal initiation in humans. Diabetes 50(8): 1714-1719. doi:10.2337/diabetes.50.8.1714 https://doi.org/10.2337/diabetes.50.8.1714
5. Damasio H, Grabowski T, Frank R, Galaburda AM, Damasio AR (1994). The return of Phineas Gage: clues about the brain from the skull of a famous patient. Science 264(5162): 1102-1105. doi:10.1126/science.8178168 https://doi.org/10.1126/science.8178168
6. Dickerson SS, Kemeny ME (2004). Acute stressors and cortisol responses: a theoretical integration and synthesis of laboratory research. Psychological Bulletin 130(3): 355-391. doi:10.1037/0033-2909.130.3.355 https://doi.org/10.1037/0033-2909.130.3.355
7. Hasselmo ME (2006). The role of acetylcholine in learning and memory. Current Opinion in Neurobiology 16(6): 710-715. doi:10.1016/j.conb.2006.09.002 https://doi.org/10.1016/j.conb.2006.09.002

8. Kandel ER, Koester JD, Mack SH, Siegelbaum SA (eds) (2021). Principles of Neural Science. 6th edition. McGraw Hill. ISBN 9781259642234.

9. LeDoux JE (2000). Emotion circuits in the brain. Annual Review of Neuroscience 23: 155-184. doi:10.1146/annurev.neuro.23.1.155 https://doi.org/10.1146/annurev.neuro.23.1.155

10. Miller EK, Cohen JD (2001). An integrative theory of prefrontal cortex function. Annual Review of Neuroscience 24: 167-202. doi:10.1146/annurev.neuro.24.1.167 https://doi.org/10.1146/annurev.neuro.24.1.167

11. Morton GJ, Cummings DE, Baskin DG, Barsh GS, Schwartz MW (2006). Central nervous system control of food intake and body weight. Nature 443(7109): 289-295. doi:10.1038/nature05026 https://doi.org/10.1038/nature05026

12. Papez JW (1937). A proposed mechanism of emotion. Archives of Neurology and Psychiatry 38(4): 725-743. doi:10.1001/archneurpsyc.1937.02260220069003 https://doi.org/10.1001/archneurpsyc.1937.02260220069003

13. Penfield W, Boldrey E (1937). Somatic motor and sensory representation in the cerebral cortex of man as studied by electrical stimulation. Brain 60(4): 389-443. doi:10.1093/brain/60.4.389 https://doi.org/10.1093/brain/60.4.389

14. Purves D, Augustine GJ, Fitzpatrick D, et al. (eds) (2018). Neuroscience. 6th edition. Sinauer Associates. ISBN 9781605353807.

15. Saper CB, Scammell TE, Lu J (2005). Hypothalamic regulation of sleep and circadian rhythms. Nature 437(7063): 1257-1263. doi:10.1038/nature04284 https://doi.org/10.1038/nature04284

16. Schultz W, Dayan P, Montague PR (1997). A neural substrate of prediction and reward. Science 275(5306): 1593-1599. doi:10.1126/science.275.5306.1593 https://doi.org/10.1126/science.275.5306.1593

17. Scoville WB, Milner B (1957). Loss of recent memory after bilateral hippocampal lesions. Journal of Neurology, Neurosurgery, and Psychiatry 20(1): 11-21. doi:10.1136/jnnp.20.1.11 https://doi.org/10.1136/jnnp.20.1.11

18. Squire LR (2004). Memory systems of the brain: a brief history and current perspective. Neurobiology of Learning and Memory 82(3): 171-177. doi:10.1016/j.nlm.2004.06.005 https://doi.org/10.1016/j.nlm.2004.06.005

19. Thannickal TC, Moore RY, Nienhuis R, Ramanathan L, Gulyani S, Aldrich M, et al. (2000). Reduced number of hypocretin neurons in human narcolepsy. Neuron 27(3): 469-474. doi:10.1016/S0896-6273(00)00058-1 https://doi.org/10.1016/S0896-6273(00)00058-1

20. Ulrich-Lai YM, Herman JP (2009). Neural regulation of endocrine and autonomic stress responses. Nature Reviews Neuroscience 10(6): 397-409. doi:10.1038/nrn2647 https://doi.org/10.1038/nrn2647

21. Weitzman ED, Fukushima D, Nogeire C, Roffwarg H, Gallagher TF, Hellman L (1971). Twenty-four hour pattern of the episodic secretion of cortisol in normal subjects. The Journal of Clinical Endocrinology and Metabolism 33(1): 14-22. doi:10.1210/jcem-33-1-14 https://doi.org/10.1210/jcem-33-1-14

10. How your answers and your name will be used

Before you start

Mnemosyn is a free study atlas of the human brain and spinal cord for students. It is not a clinical tool and must not be used for diagnosis or treatment. We ask specialists to check its anatomical and clinical statements before students learn from them.

What we collect

Your name, email address, optional other contact details, your speciality, optional institution and years in practice, your answers and your notes.

Why

To decide which statements are accurate and to correct those that are not. Your answers are combined with those of other reviewers into statistics, such as how many reviewers agreed with each statement.

Credit

You choose whether we mention you. If you agree, your name (and your institution if you choose) may be listed as a reviewer in the application, its documentation and its website. If you do not agree, your name is never published. Your email address and contact details are never published.

Use of your notes

We may use the substance of your corrections to rewrite our texts. We will not quote you by name without asking you first.

Where your data is kept

In a private spreadsheet owned by the project, visible only to the project team. You can ask us at any time to change your credit choice or to delete your answers by writing to alsaeedalbasi0ny@gmail.com.

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Reviewing is voluntary and unpaid. The project team stays responsible for what the application finally says; your review advises us and does not make you responsible for the content.

Thank you. Your answers will shape what students learn about the brain.